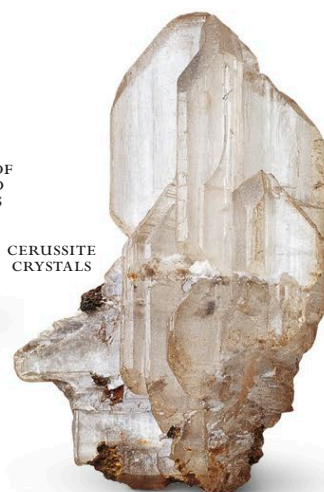


CLUSTER OF
TWINNED
CRYSTALSCERUSSITE
CRYSTALS**CERUSSITE**

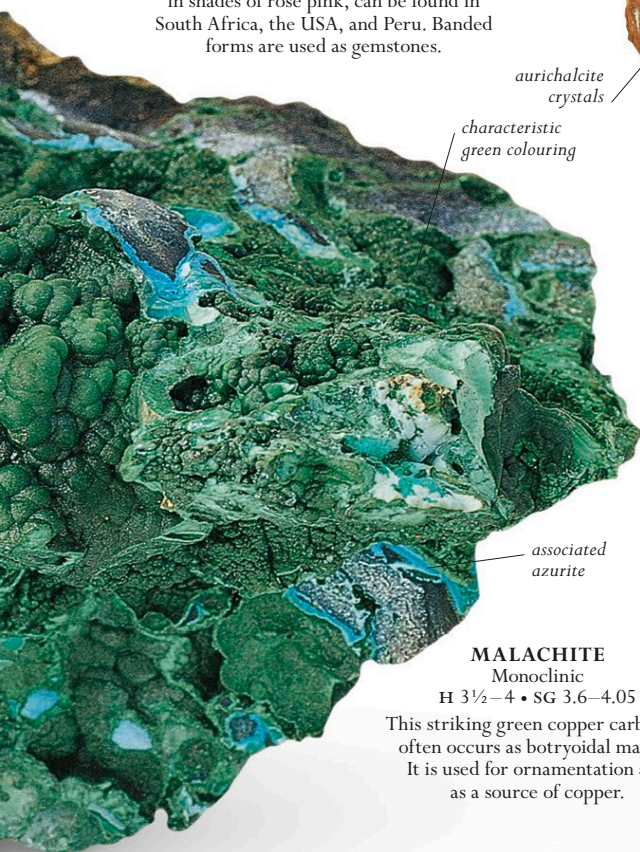
Orthorhombic • H 3–3½ • SG 6.53–6.57

A carbonate of lead occurring where lead-bearing veins have been altered, cerussite is the most common lead ore after galena.

rhombohedral
crystals**RHODOCHROSITE**

Trigonal • H 3½–4 • SG 3.7

Fine crystals of this manganese carbonate, in shades of rose pink, can be found in South Africa, the USA, and Peru. Banded forms are used as gemstones.

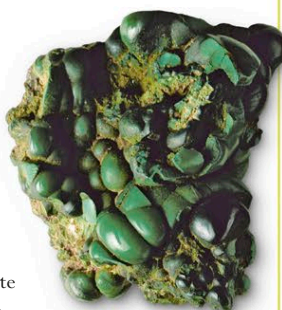
aurichalcite
crystalscharacteristic
green colouringassociated
azurite**MALACHITE**Monoclinic
H 3½–4 • SG 3.6–4.05

This striking green copper carbonate often occurs as botryoidal masses. It is used for ornamentation and as a source of copper.

**AURICHALCITE**

Monoclinic • H 1–2 • SG 3.96

A blue- or green-coloured zinc copper carbonate hydroxide, aurichalcite forms in the oxidized zones of zinc and copper deposits.

MALACHITE ON
CHRYSOCOLLABOTRYOIDAL
MALACHITE**BORATES**

Borates occur when metallic elements combine with the borate radical (BO_3). There are over 100 borate minerals, the most common being borax, kernite, ulexite, and colemanite. Borates tend to be pale coloured, relatively soft, and have low specific gravity. Many occur in evaporite sequences where saline waters have dried out and minerals are then precipitated among layers of sedimentary rocks.

**KERNITE**

Monoclinic • H 2½ • SG 1.91

A colourless or white sodium borate hydrate, kernite has less water than borax. The two minerals often occur together.

**BORACITE**Orthorhombic
H 7–7½ • SG 2.91–3.1

Crystals of this magnesium borate chloride are pale green or white, with a vitreous lustre and high hardness. Boracite occurs in salt deposits.

translucent,
prismatic
crystals**ULEXITE**

Triclinic • H 2½ • SG 1.95

A hydrated sodium calcium borate hydroxide, ulexite's white, fibrous crystals transmit light down their length. It has uses similar to borax.

This hydrated calcium borate hydroxide forms when saline water evaporates. It was the main source of boron until the discovery of kernite.

COLEMANITE

Monoclinic • H 4½ • SG 2.42

**HOWLITE**

Monoclinic • H 3½–6½ • SG 2.6

Howlite is a calcium borosilicate hydroxide. It generally forms as chalky, rounded masses.

This chalky-white sodium borate hydrate, borax has many applications, including in medicines, washing powders, glasses, and textiles.

BORAX

Monoclinic • H 2–2½ • SG 1.71

A chalky-white sodium borate hydrate, borax has many applications, including in medicines, washing powders, glasses, and textiles.

NITRATES

Nitrates are a small group of compounds where metallic elements combine with the nitrate radical (NO_2). These minerals are usually very soft and have low specific gravity. Many dissolve readily in water and they only rarely occur as crystals. They are generally confined to arid regions, forming coatings on the land surface that often cover wide areas. Commercially, nitrates can be used as fertilizers or explosives.

**NITRATINE**

Trigonal • H 1½–2 • SG 2.26

This sodium nitrate typically occurs as crusts on the ground surface in arid regions, especially in Chile. It is white, grey, brown, or yellow.